

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	Saee B	Reg. No.:	192524023
Section / Batch:	AI and DS / 2025	Date:	01/08/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I Saee B (192524023) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B. S

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				

Total Marks (100):

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Brute Force Maximum Subarray Problem

Given the array: $A = [-2, 3, 4, -1, -2, 1, 5, -3]$

To Find the maximum subarray sum using Brute Force Approach.

Method

The brute force method checks every possible contiguous subarray, calculates its sum, and keeps track of the largest sum found.

Sub-array Calculations

Sub array	Sum
$[-2]$	-2
$[-2, 3]$	1
$[-2, 3, 4]$	5
$[-2, 3, 4, -1]$	4
$[3]$	3
$[3, 4]$	7
$[3, 4, -1]$	6
$[3, 4, -1, -2]$	4
$[3, 4, -1, -2, 1]$	10
$[3, 4, -1, -2, 5]$	10
$[4]$	4
$[4, -1]$	3

[4, -1, -2, 1, 5]	7
[5]	5
[5, -3]	2

All 36 possible subarrays are examined.

Number of Comparisons

For any array of $n=5$

$$\text{Number of sub arrays} = \frac{n(n+1)}{2} = \frac{5 \times 6}{2} = 15$$

Hence, 36 comparisons (maximum updates / checks) are made.

Maximum sum: * Maximum Subarray: [3, 4, -1, -2, 1, 5]

* Maximum Sum = 10

$$\text{Verification: } 3 + 4 - 1 - 2 + 1 + 5 = 10$$

Time Complexity: * Best Case: $O(n^2)$

* Average Case: $O(n^2)$

* Worst Case: $O(n^2)$

If every subarray sum is recomputed separately, it becomes $O(n^3)$

Interpretation

The brute force approach guarantees the correct answer by checking every possible subarray. However, it is inefficient for large datasets because the number of subarrays grows rapidly, making the execution time high.

Linear Search - Unsuccessful search in unsorted Data

Problem : Given the unsorted dataset: (16, 28, 39, 47, 58, 69, 74)

Search for the element 100 using Linear Search

Method : Linear Search compares the target element with each element in the list from left to right until it is found or the list ends.

Comparison	Array Element	Result
1	16	$100 \neq 16$
2	28	$100 \neq 28$
3	39	$100 \neq 39$
4	47	$100 \neq 47$
5	58	$100 \neq 58$
6	69	$100 \neq 69$
7	74	$100 \neq 74$

After checking all elements, the search ends.

Total Comparisons

* Total elements = 7

* Total Comparisons = 7

Result :

Element 100 is not present in the dataset

Verification

All seven elements were checked and none matched the target value.

Time Complexity

* Best Case: $O(1)$ (element found at first position)

* Average Case $O(n)$!

* Worst Case $O(n)$! (element is last or not present)

Interpretation

Linear Search is simple and works on both sorted and unsorted data. In an unsuccessful search, every element must be checked, making it inefficient for large datasets. For this example all 7 comparisons were required before concluding that 100 is absent.